Lecture 16 recap: Fourier series

- The *Fourier series* of a function $f(x)$ for $(-L, L]$ is

$$f(x) \stackrel{?}{=} \frac{1}{2}a_0 + \sum_{n=1}^{\infty}\Big[a_n \cos(k_n x) + b_n \sin(k_n x)\Big] \qquad \text{where} \qquad k_n = \frac{n\pi}{L}$$

- The coefficients are given by Euler-Fourier formulae

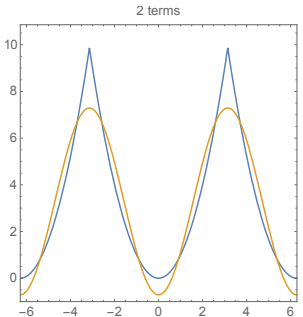
$$a_n = \langle \cos(k_n x), f(x)\rangle = \frac{1}{L}\int_{-L}^{L} f(x)\cos(k_n x)\,\mathrm{d}x\,, \qquad n = 0, 1, 2, \ldots$$

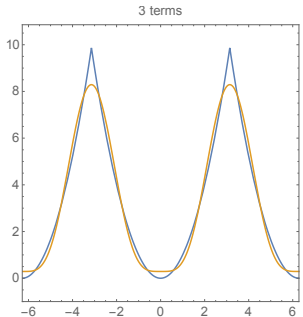
$$b_n = \langle \sin(k_n x), f(x)\rangle = \frac{1}{L}\int_{-L}^{L} f(x)\sin(k_n x)\,\mathrm{d}x\,, \qquad n = 1, 2, 3, \ldots$$

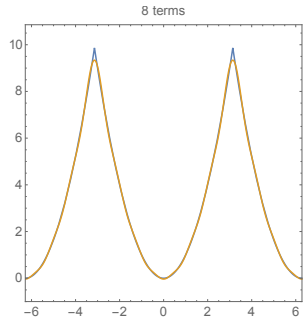
- We denote the periodic extension of $f(x)$ as $F(x)$

Fourier series of $f(x) = x^2$ on $(-\pi, \pi]$

$$F(x) \stackrel{?}{=} \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos(nx)$$

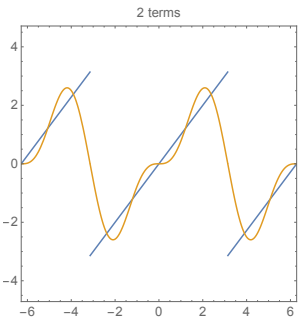


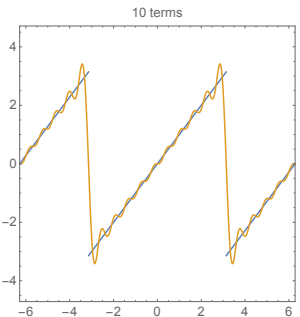


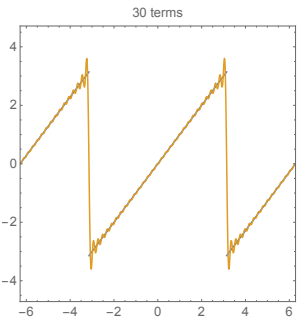


Fourier series of $f(x) = x$ on $(-\pi, \pi]$

$$F(x) \stackrel{?}{=} -2 \sum_{n=1}^{\infty} \frac{(-1)^n}{n} \sin(nx)$$







Lecture 17

In Lecture 17, we will:

1. Outline conditions for pointwise and uniform convergence of Fourier series

Pointwise convergence of a Fourier series (Dirichlet's sufficient condition)

Let f be piecewise continuously differentiable on $(-L, L]$, then:

- The Fourier series of f exists, i.e., the Euler-Fourier integrals all converge.
- The Fourier series converges to the periodic extension, $F(x)$, where it is continuous.
- For any x where $F(x)$ is discontinuous, the Fourier series instead converges to

$$\frac{1}{2}\left(\lim_{w \to x^-} F(w) + \lim_{w \to x^+} F(w)\right)$$

  This is similar to Laplace transforms!

$f(x) = x^2$ and $f(x) = x$ piecewise $C^1 \Rightarrow$ pointwise convergence of Fourier series

Exercise 4.5: Evaluate the series $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$

This is a neat application of the pointwise convergence of a Fourier series.

From Ex. 4.3 + pointwise convergence

$$f(x) = x^2 = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}\cos(nx) \quad \Rightarrow \quad f(0) = 0 = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$$

$$\Rightarrow \qquad \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} = -\frac{\pi^2}{12}$$

Recap (Topic 2): Uniform convergence of series / Weierstrass test

- The series $\sum_{n=1}^{N} w_n(t)$ converges uniformly to $W(t)$ on D if

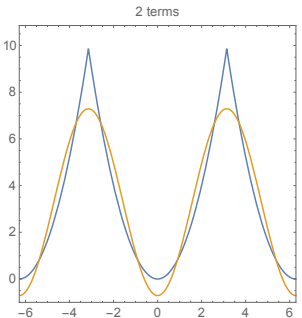
$$\lim_{N\to\infty} \max_{t\in D} \left| \sum_{n=1}^{N} w_n(t) - W(t) \right| = 0$$

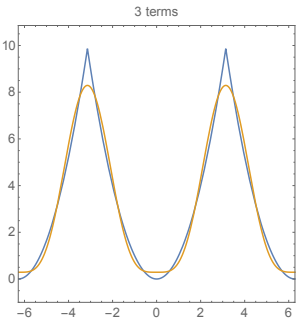
- Let $w_j(t)$, $j = 1, 2, 3, \ldots$ be functions on domain D. If $\max_{t\in D} |w_j(t)| \leqslant a_j$ for some $a_j \in \mathbb{R}$ such that $\sum_{j=1}^{\infty} a_j$ converges then $\sum_{j=1}^{\infty} w_j(t)$ converges *uniformly* on D.
- A uniformly convergent series 'smoothly' converges to the original function
- A uniformly convergent series can be integrated term by term
- But need extra condition for term by term differentiation

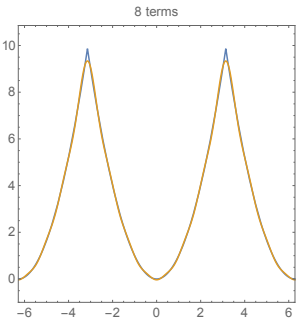
Exercise 4.6: Does the Fourier series of x^2 converge uniformly?

Recall

$$F(x) = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos nx$$







Consider the series $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^2}$

We know from MAST10006 that this series converges.

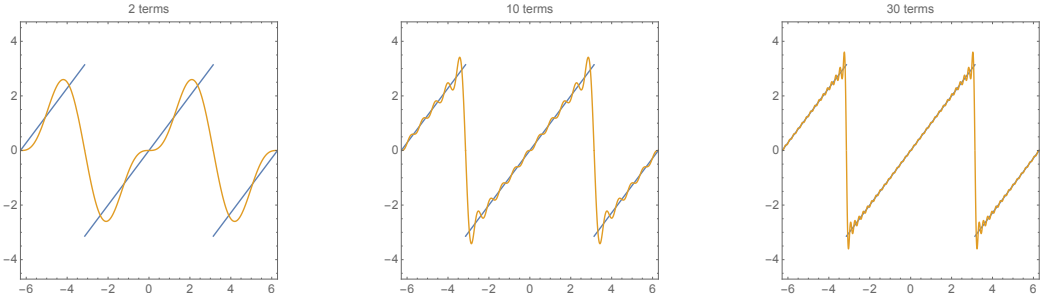
Now as $\left|\dfrac{(-1)^n}{n^2}\cos nx\right| \leqslant \dfrac{1}{n^2}$ for all x and $n \geqslant 1$, and so the Fourier series of x^2 converges uniformly.

Theorem

If f' is piecewise continuous on $x \in (-L, L]$, and the periodic extension of f is continuous on $\mathbb{R}$, then the Fourier series of f converges *uniformly* to the periodic extension of f.

- If f satisfies the conditions for pointwise convergence of its Fourier series but the periodic extension of f is discontinuous, then the convergence will NOT be uniform.

- The Fourier series of a discontinuous function overshoots and "rings" on either side of the discontinuity, which is known as *Gibbs phenomenon*



Case study 6: creating an electronic music synthesiser

